

Lecture 9 Live Problems

Power function growth. From difference to derivative.

Points toward lab manual pages: **D1, D2, D3, RT1, RT2** (section L2 S2)

1. A mold spot. The radius of a mold spot is $R(t) = 5t$. Units are millimeters (and days).

- (a) How large is the mold spot on day 6?
- (b) What is its area A , as a function of time? What is the area on day 6?
- (c) Estimate the number of mold spores P , given each one is 10 microns across. How many mold spores fit in 1 mm^2 ? Express the answer as a function $P(t) = \underline{\hspace{2cm}}$
- (d) How many mold spores on day 6? What is the rate of spore population growth on day 6?

2. The derivative of $P(t) = t^2$.

(a) What is the rate of change $\Delta P / \Delta t$ on the time interval $[2, 3]$? _____

(b) Estimate the instantaneous rate of change dP/dt at $t = 2$: _____

This is called the *derivative* of P at $t = 2$, or $P'(2)$. Say P prime of 2." Why two notations? Leibniz created the dP/dt notation, and Newton the $P'(t)$ notation.

3. The derivative of $Q(t) = t^{0.5}$.

(a) What is the rate of change $\Delta Q / \Delta t$ on the time interval $[4, 9]$? _____

(b) Estimate the instantaneous rate of change dQ/dt at $t = 4$: _____

This is the derivative of Q at $t = 4$, or $Q'(4)$.

4. The Power Rule (Newton and Leibniz, mid/late 1600s). If $F(t) = C t^p$, then the derivative at time t is $\frac{dF}{dt} = C p t^{p-1}$. Examples:

$A(t) =$	$dA/dt =$
$B(t) =$	$dB/dt =$
$C(t) =$	$dC/dt =$
$E(t) =$	$dE/dt =$
$F(t) =$	$dF/dt =$

Now remember our function $Q(t) = t^{0.5}$. Then $dQ/dt =$ _____ and at $t = 4$, $dQ/dt =$ _____.